

Homework IV

Typed graphs over a fixed type graph form a category

1 Prerequisites & notation

Throughout the exercise we work with *directed graphs*. A graph is a quadruple $G = (E, V, s, t)$ where

$$s, t : E \longrightarrow V$$

are the source and target maps. A *graph morphism* $f : G_1 \rightarrow G_2$ is a pair $f = (f_E, f_V)$ of functions $f_E : E_1 \rightarrow E_2$, $f_V : V_1 \rightarrow V_2$ such that $s_2 \circ f_E = f_V \circ s_1$ and $t_2 \circ f_E = f_V \circ t_1$.

Fix once and for all a *type graph*

$$\mathbf{TG} = (E_{\mathbf{TG}}, V_{\mathbf{TG}}, s_{\mathbf{TG}}, t_{\mathbf{TG}}).$$

All typing data in what follows refer to this graph.

2 Typed graphs and their morphisms

Definition 2.1 (Typed graph). A *graph typed over* $\mathbf{TG}$ (short: a *typed graph*) is a pair (G, type) where

$$\text{type} = (\text{type}_E, \text{type}_V) : G \rightarrow \mathbf{TG}$$

is a graph morphism. Explicitly,

$$s_{\mathbf{TG}} \circ \text{type}_E = \text{type}_V \circ s, \quad t_{\mathbf{TG}} \circ \text{type}_E = \text{type}_V \circ t.$$

Definition 2.2 (Morphisms of typed graphs). Given typed graphs (G_1, type_1) and (G_2, type_2) , a *typed-graph morphism*

$$f = (f_E, f_V) : (G_1, \text{type}_1) \longrightarrow (G_2, \text{type}_2)$$

is a graph morphism $G_1 \rightarrow G_2$ that *preserves type*, i.e.

$$\text{type}_{1,E} = \text{type}_{2,E} \circ f_E, \quad \text{type}_{1,V} = \text{type}_{2,V} \circ f_V.$$

$$\begin{array}{ccc} G_1 & \xrightarrow{f} & G_2 \\ \text{type}_1 \downarrow & & \downarrow \text{type}_2 \\ \mathbf{TG} & \xlongequal{\quad} & \mathbf{TG} \end{array}$$

3 Main result

Theorem 3.1. *The class of typed graphs over $\mathbf{TG}$, together with typed-graph morphisms, forms a category, denoted $\mathbf{Graph}_{/\mathbf{TG}}$.*

Proof. We must supply composition of morphisms and verify the four category axioms.

(1) Composition is well-defined Let

$$(G_1, \text{type}_1) \xrightarrow{f} (G_2, \text{type}_2) \xrightarrow{g} (G_3, \text{type}_3)$$

be typed-graph morphisms. The ordinary composition $g \circ f$ is a graph morphism $G_1 \rightarrow G_3$. Type preservation is inherited:

$$\text{type}_{3,E} \circ (g_E \circ f_E) = (\text{type}_{3,E} \circ g_E) \circ f_E = \text{type}_{2,E} \circ f_E = \text{type}_{1,E},$$

and similarly on vertices. Hence $g \circ f$ is again a typed-graph morphism. Closure holds.

(2) Associativity Composition is defined using the underlying category **Graph**, which is associative; therefore $(h \circ g) \circ f = h \circ (g \circ f)$ as typed-graph morphisms.

(3) Identities For every object (G, type) the identity graph morphism $\text{id}_G = (\text{id}_E, \text{id}_V)$ satisfies $\text{type}_E = \text{type}_E \circ \text{id}_E$ and $\text{type}_V = \text{type}_V \circ \text{id}_V$, hence qualifies as a typed-graph morphism $\text{id}_{(G, \text{type})}$.

(4) Identity laws Because identities and composition come from **Graph** they obey $f \circ \text{id} = f = \text{id} \circ f$.

All four axioms are satisfied, so **Graph**_{/TG} is indeed a category. □

4 Two corollaries (optional, but nice to mention)

Corollary 4.1 (Labelled graphs). *If TG is a “star” whose vertices/loops encode labels, then typed graphs over TG coincide with ordinary labelled graphs.*

Corollary 4.2 (Forgetful functor). *There is a faithful functor*

$$U: \mathbf{Graph}_{/TG} \longrightarrow \mathbf{Graph}, \quad (G, \text{type}) \mapsto G, \quad f \mapsto f,$$

which simply forgets the typing. In many contexts U admits a left adjoint (free typing) and, under mild finiteness conditions, also a right adjoint (constant typing).

Remark. The category constructed above is the *slice category* **Graph**/TG; this homework is a pleasant concrete case study of that general construction.

END OF SOLUTION